

Composite Liner Tutorial

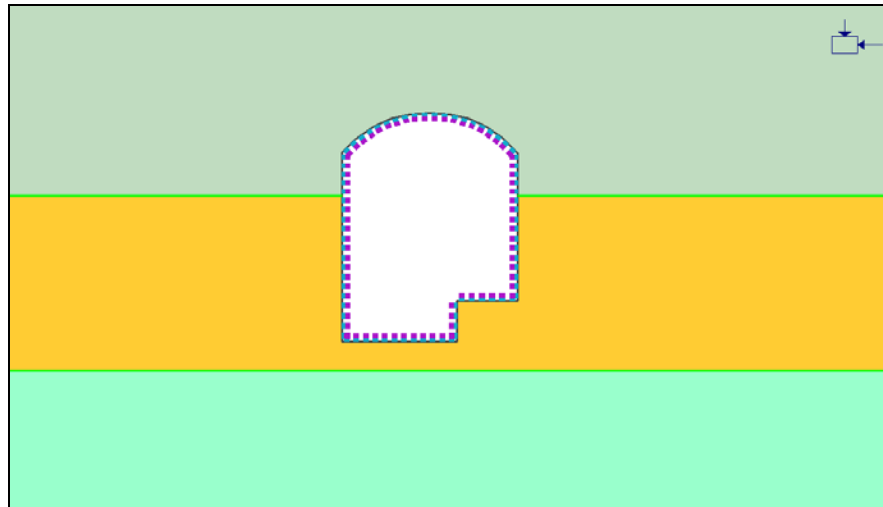
This tutorial describes the support of a tunnel using composite liners. A Composite Liner in *Phase2*, is a liner which may consist of multiple layers of material. The different layers of a Composite Liner may have different material properties, and may be applied at different stages. A joint may also be included in the Composite Liner, which will exist between the rock mass and the first liner layer.

There are a total of four stages in the simulation. The tunnel will be excavated in three stages. After each excavation, shotcrete is added for support. A layer of concrete is added one stage later on top of the shotcrete layer to form a composite liner. The installation of the concrete liner is finished in the fourth stage.

Topics Covered

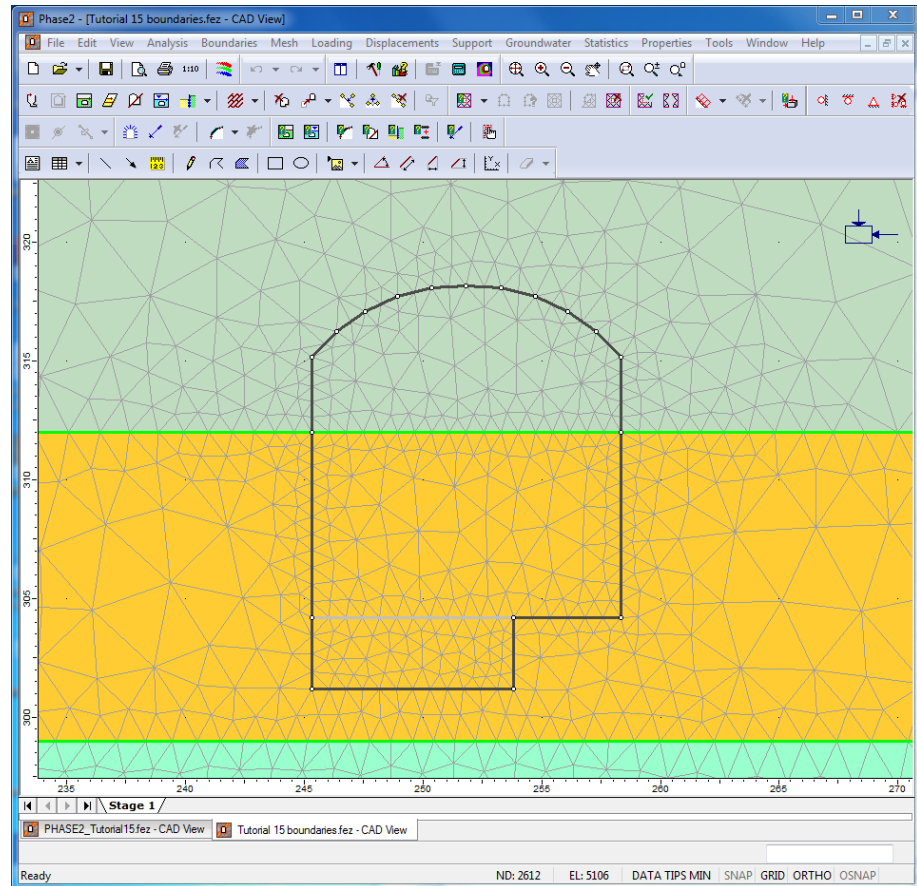
- Composite Liners
- Staged liner installation
- Selection Window
- Selection Filter
- Graph composite liner data
- Show values

Geometry



Model

Start the *Phase2* Model program. To save typing in boundaries and material properties we will start with an existing model and then excavate it and add support. Select **File > Recent Folders > Tutorials Folder** and open the file **Tutorial 15 boundaries.fez**. Zoom into the excavation and the model should look like this.



Project Settings

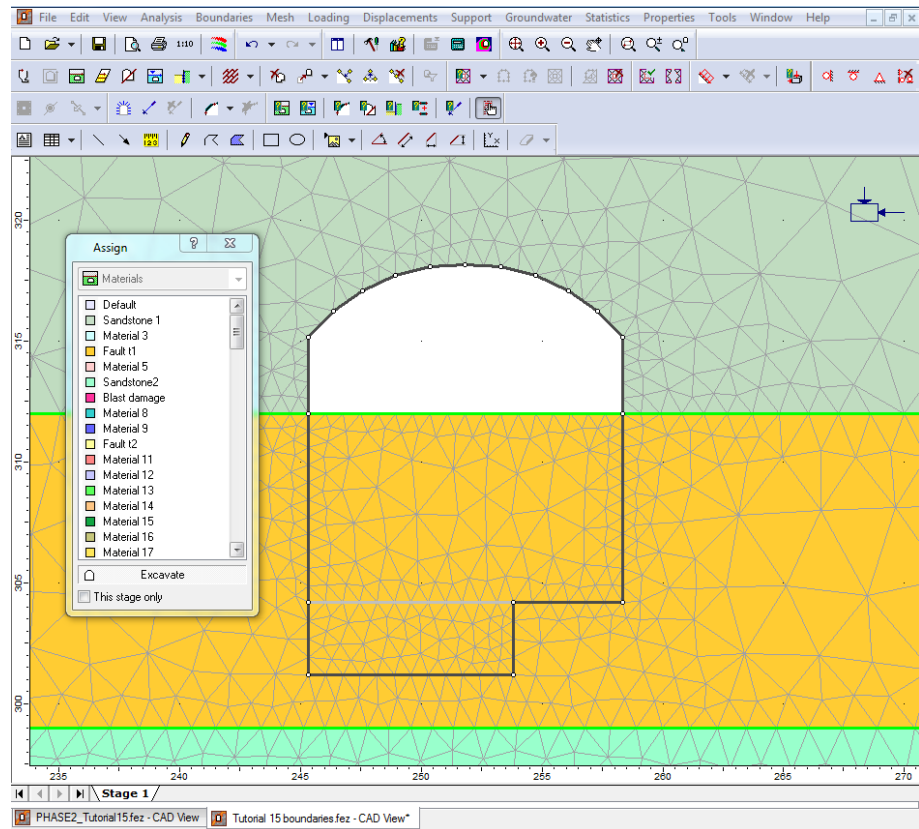


Click the Project Settings button on the toolbar. Change the number of stages to 4. Close the dialog by clicking OK.

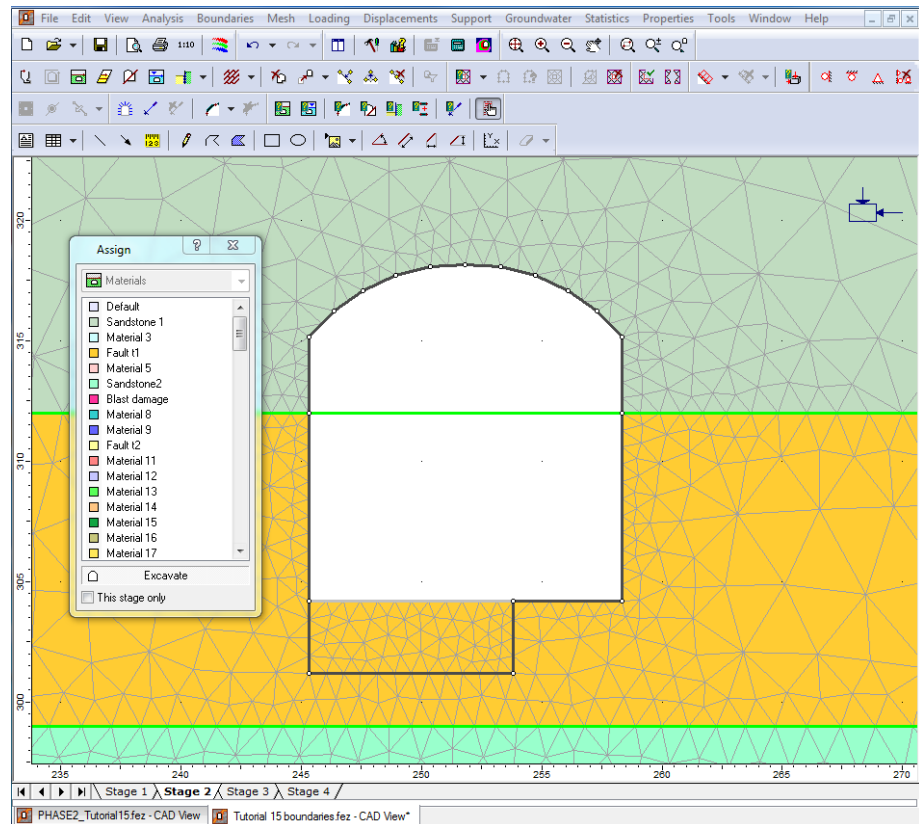
Excavation



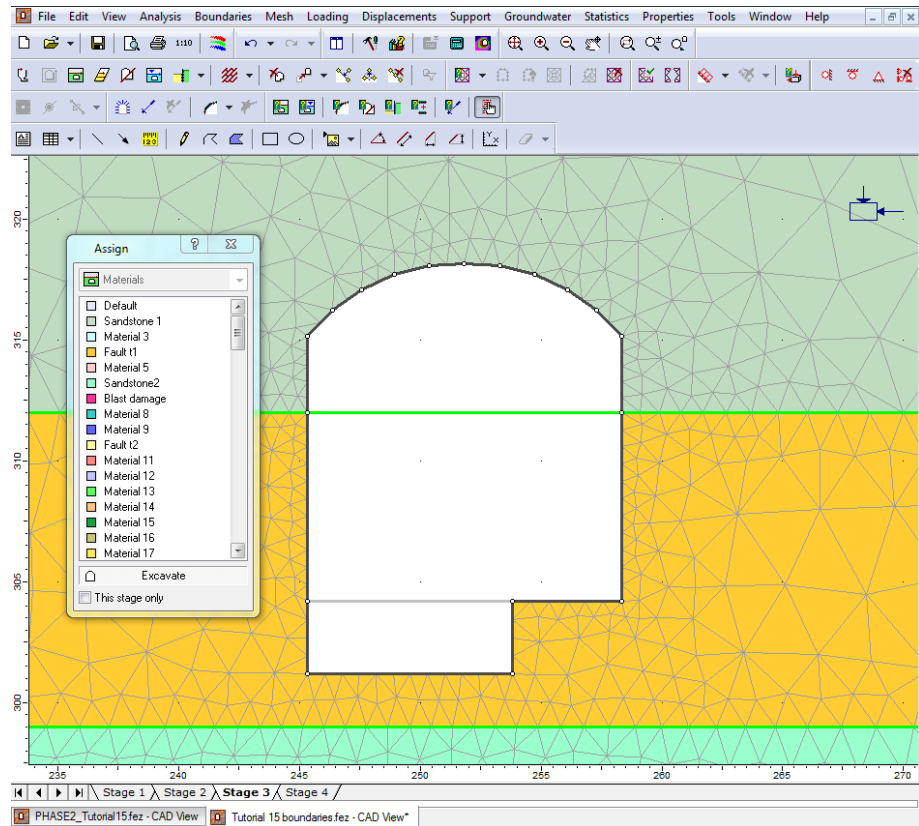
Click on the Stage 1 tab. Select the Assign Properties button on the toolbar. From the Assign Properties dialog, select Excavate. Click anywhere within the upper tunnel section. Your screen should look like the following:



In Stage 2 we will excavate the middle section of the tunnel. Click the Stage 2 tab. Click anywhere within the middle tunnel section. (Note that the Excavate option you selected from the previous step is still active.) Your screen should look like the following:



In Stage 3 we will excavate the lower section of the tunnel. Click the Stage 3 tab. Click anywhere within the lower tunnel section.



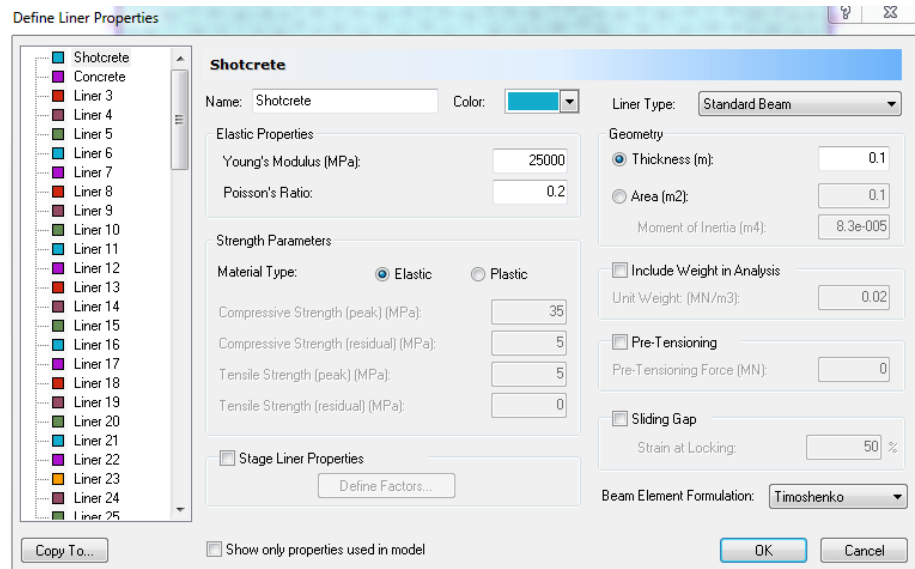
We are now finished excavating the tunnel. Close the Assign dialog.

Composite Liner Properties

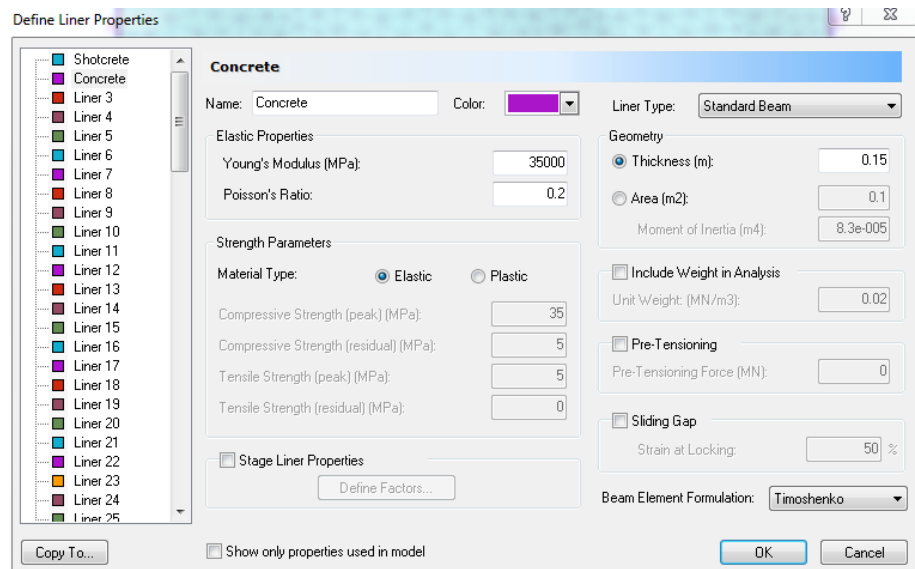
Composite Liners are made up of two or more single liners and possibly a joint (slip element) between the rock and the first liner layer. Before specifying the properties of the composite liner, we will set the properties of the individual liner layers.



Click the Define Liner Properties button on the toolbar. In our model, the composite liner will be made up of a shotcrete layer and a concrete layer. Change the Name of Liner 1 to Shotcrete and change the Young's modulus to 25000 MPa. The dialog should look like this.



We will now define the properties of the second layer of the composite liner. Click on the tab for Liner 2. Change the name to Concrete, the Young's Modulus to 35000 MPa, and the Thickness to 0.15 m. The dialog should look like this.



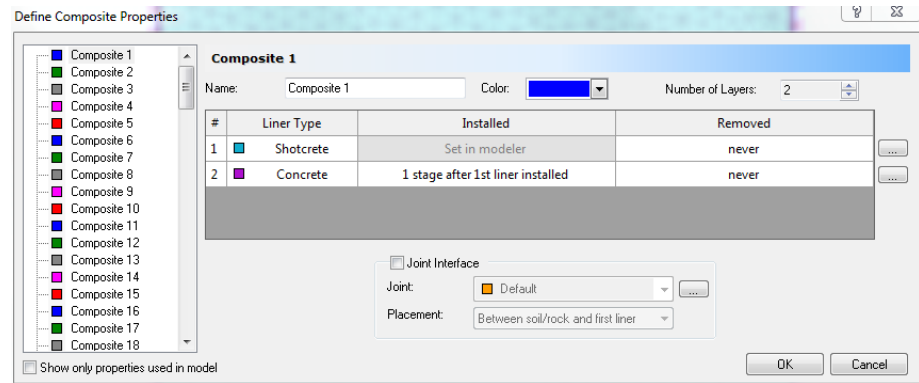
Click OK to close the dialog.



Now we can set up the composite liner. Click the Define Composite Properties button on the toolbar.

Our composite liner is to be made up of the shotcrete layer and concrete layer we have already defined. Set the Number of Layers to 2 and ensure that the **First Liner** is set to Shotcrete and the **Second Liner** is set to Concrete.

In our model we want the second layer to be placed one stage after the first. This means that if, for example, the first shotcrete layer of the composite liner is installed at Stage 2, the second concrete layer will be applied at Stage 3. To accomplish this, go to the drop down menu in the Installed column next to the word **Concrete**. Select “1 stage after 1st liner installed” from this menu. Your dialog should now look like this.



Click OK to close the dialog.

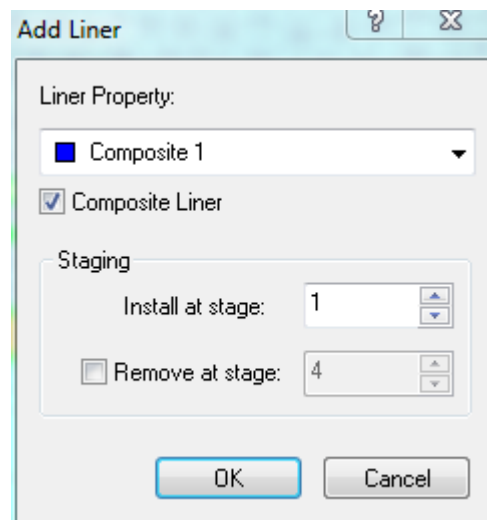
NOTE: You can also specify that slip may occur between the rock and the composite liner by choosing “2 liners (with slip)” from the Composite Type drop down menu, but we will not be doing that for this example.

Add Support

In this model, we will add the entire liner at Stage 1 and then stage the installation later.

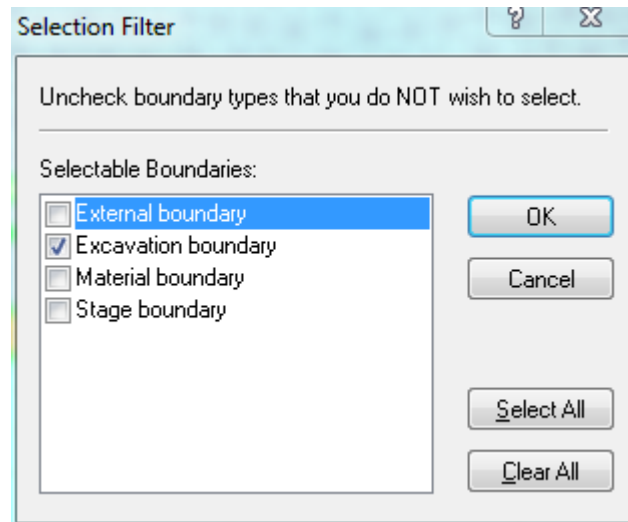


To add the composite liner first go to Stage 1. Click the Add Liner toolbar button. In the Add Liner dialog, click the Composite Liner checkbox so that the Liner Property is Composite 1. The value for **Install at stage** should be 1 as shown.

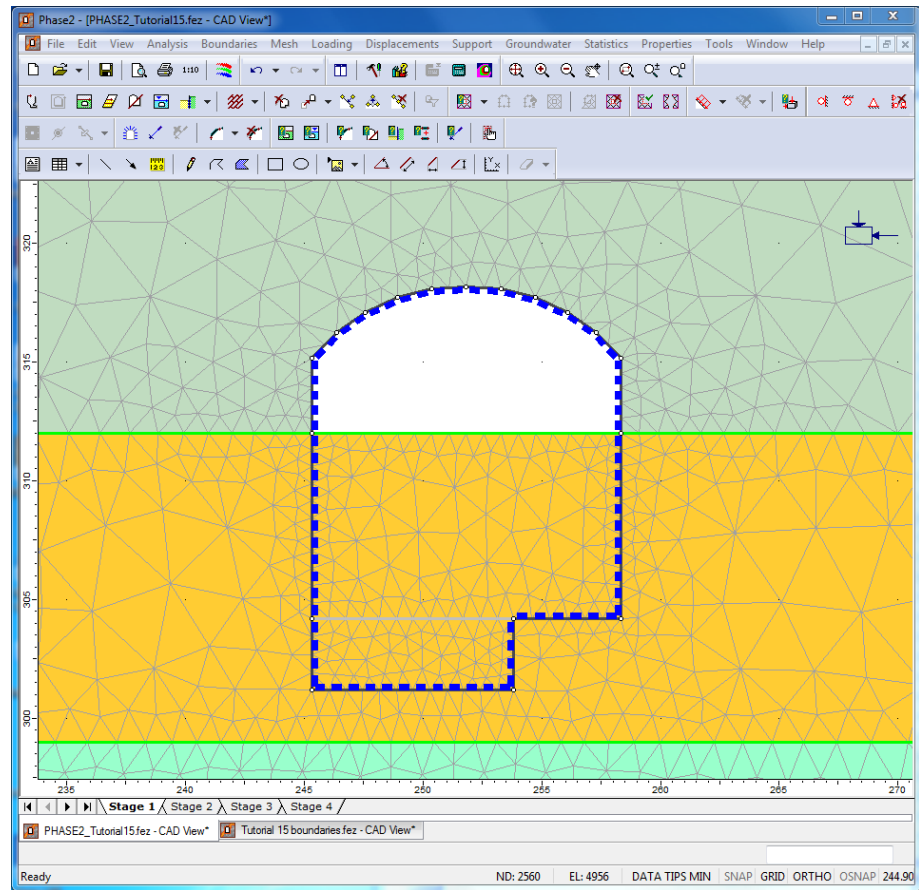


Click OK to close the dialog. Now select all of the segments that make up the large tunnel. Be sure to select the parts of the tunnel that are unexcavated at Stage 1 – we will deal with the staging later. Hit Enter to finish selection.

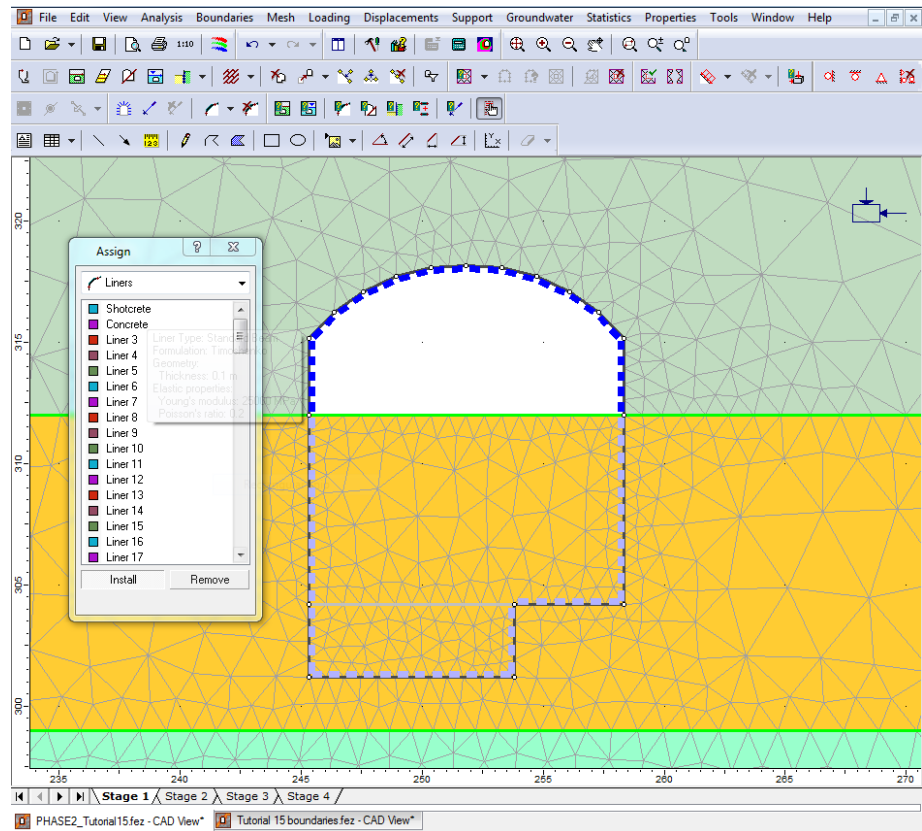
TIP: You can easily select all the sections of the tunnel using a Selection Window. Hold down the left mouse button and drag a window to encompass the entire tunnel. If you do this, you will find you have selected the internal stage and material boundaries as well as the excavation boundary. To prevent this from happening you can use a Selection Filter. Before drawing your Selection Window, right click and choose Selection Filter. Deselect everything except Excavation Boundary as shown. Now click OK and draw the Selection Window.



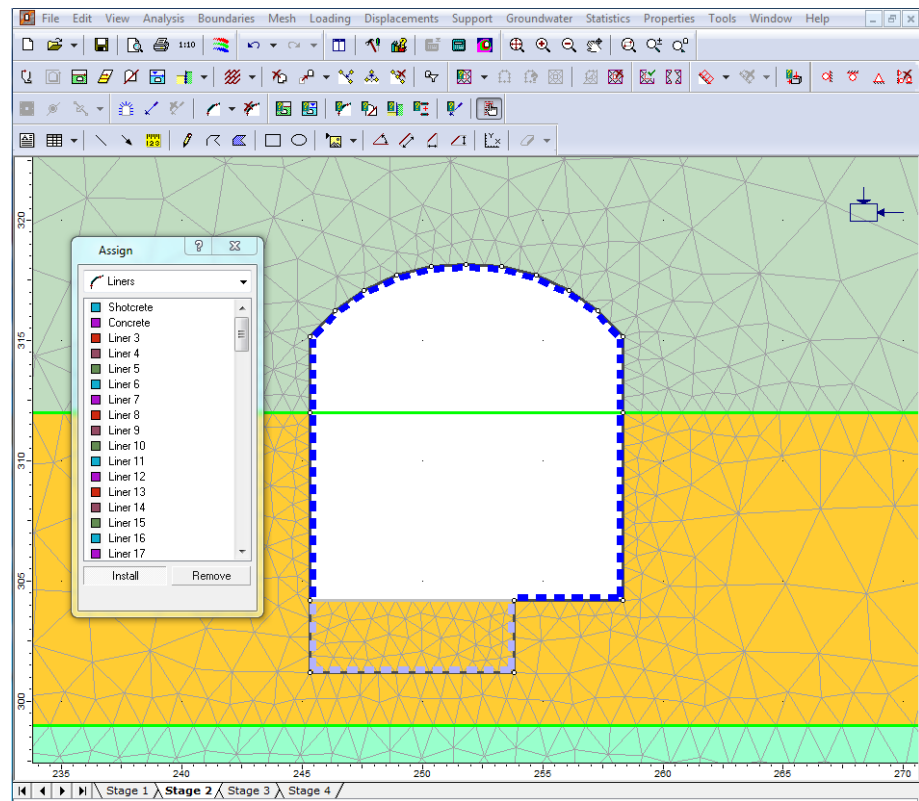
Your model should look like this.



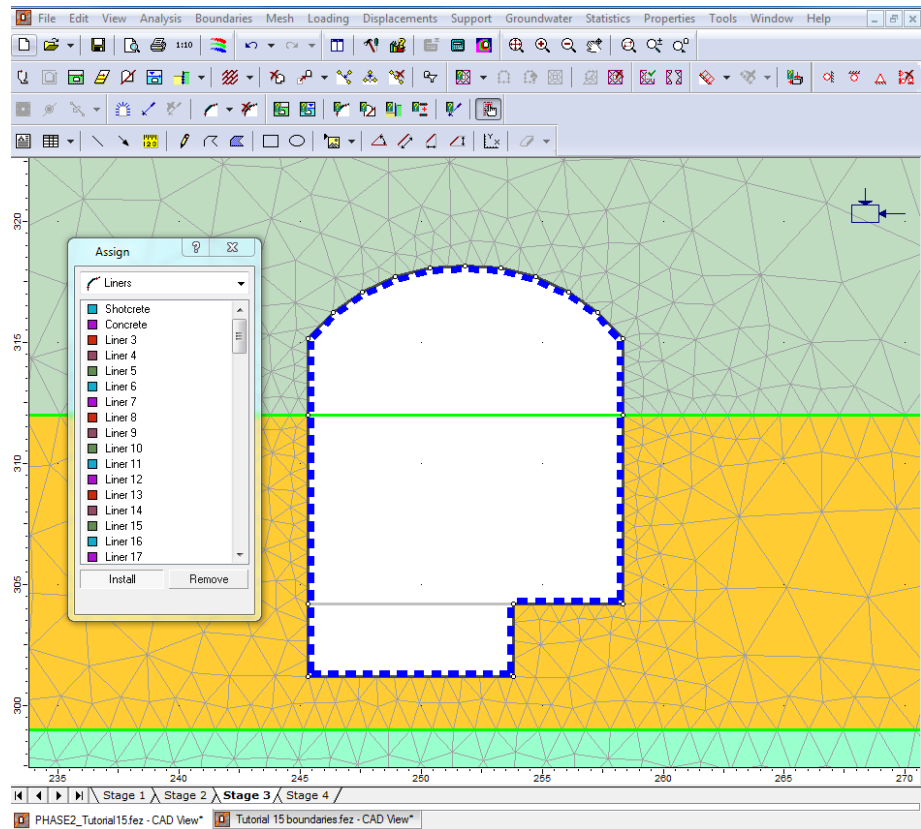
To assign the composite liner installation to the correct stages, first click the Assign Properties button on the toolbar. Select Composite Liners from the drop down menu at the top. Now click the Remove button and click on all sections of the tunnel that have not been excavated in Stage 1. Hit Enter. The sections you selected will turn grey and your model should look like this.



Now select the Stage 2 tab. Click the Install button on the Assign dialog. Select the liner segments that become exposed at this stage of excavation (this includes the two exposed vertical walls and floor ledge) and hit the enter key.



Click the Stage 3 tab and select the additional liner segments that become exposed at this stage of excavation. Hit the enter key. Close the Assign dialog. The model for Stage 3 should now look like this.



NOTE: in Stage 4, the concrete layer will be installed on the bottom section of the tunnel since the second layer of the composite liner is installed one stage after the first (shotcrete) layer.

Save your model by choosing Save As from the File menu and give the file a different name.

Compute



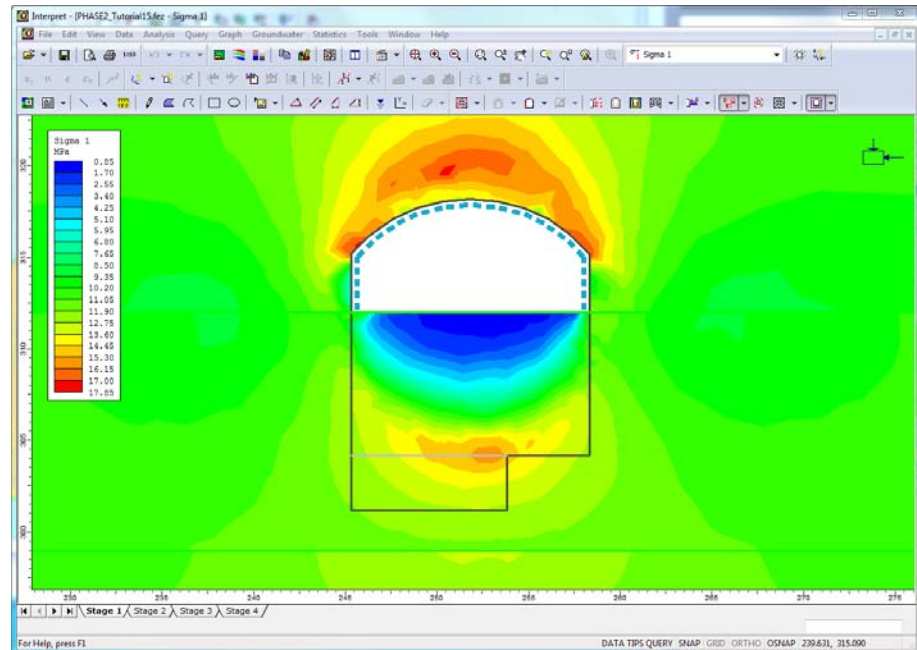
Run the model by pressing the Compute button on the toolbar. The analysis should take under a few minutes to run.

Once the model has finished computing (Compute dialog closes), click the Interpret button to view the results.

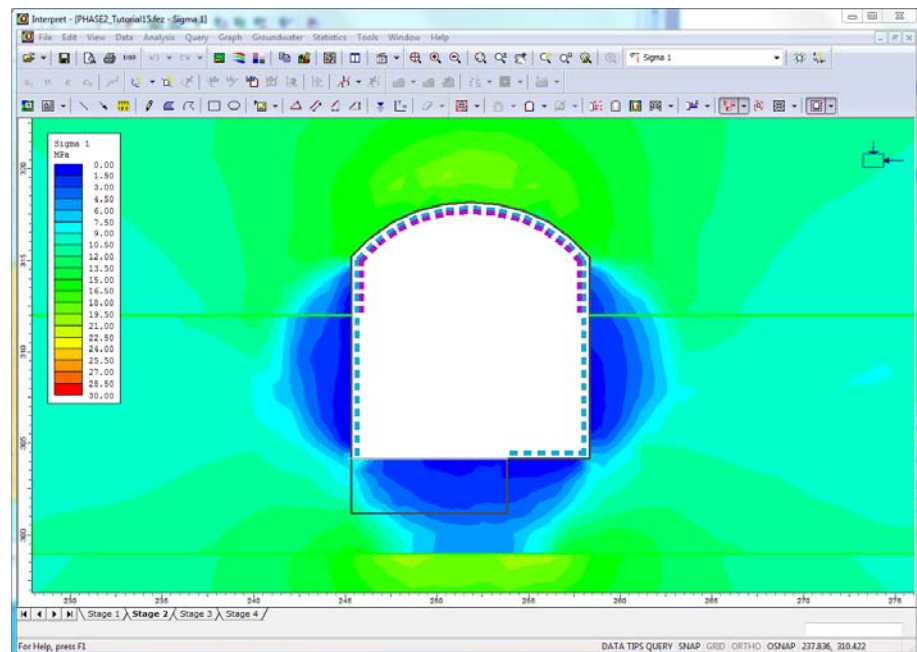
Interpret



After you select the Interpret option, the Interpret program starts and reads the results of the analysis. Zoom into the excavation using the Zoom Excavation button. You should see a screen similar to the following that shows the maximum compressive stress for Stage 1.



You can see the installation of the shotcrete liner around the excavation (marked as light blue rectangles). Select the Stage 2 tab.

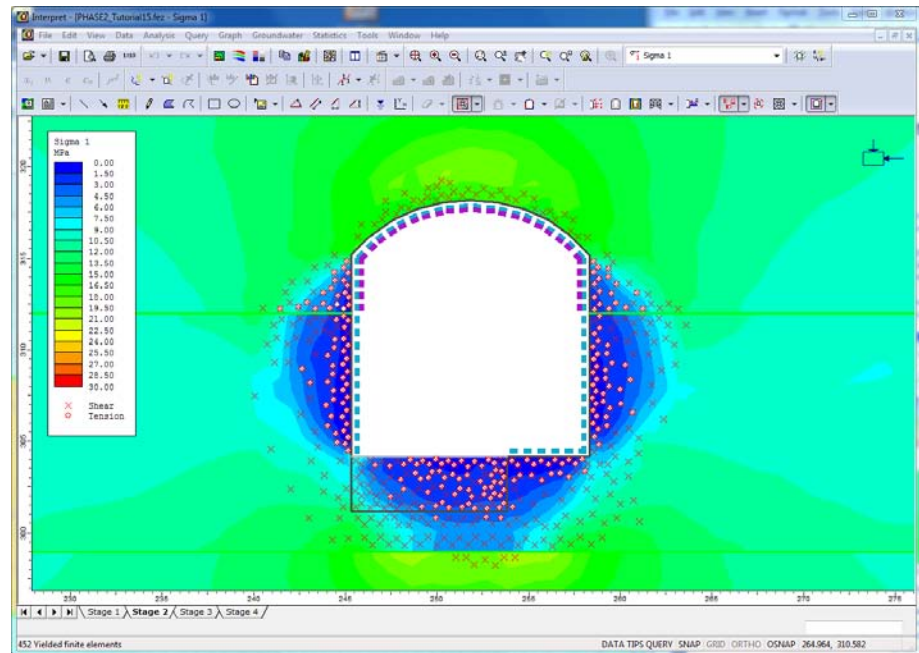


Purple rectangles show a second layer of support has been added (concrete) to the top of the tunnel that was excavated in Stage 1.

This plot shows low stresses directly below the tunnel with high stresses shed into the sandstone layer further below. This suggests failure has occurred below the tunnel.



To observe the failure in the rock, click on the Display Yielded Elements button. The plot should now look like this.



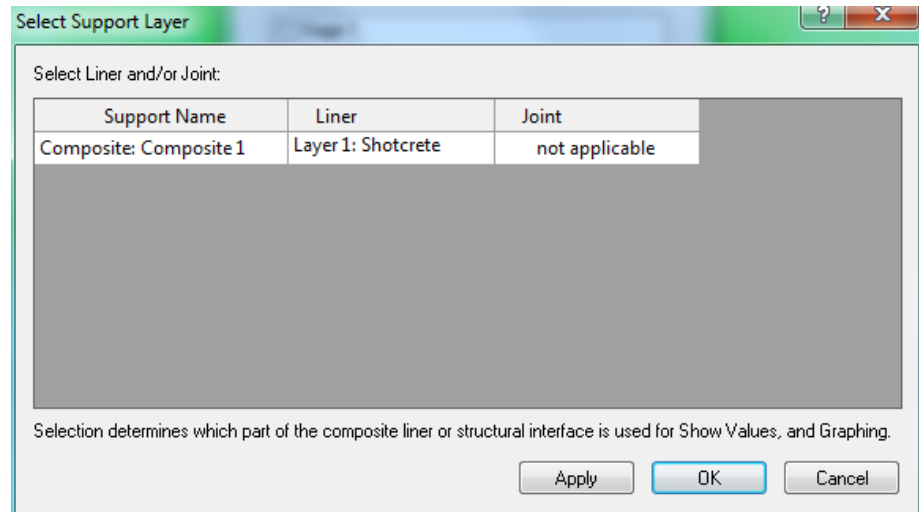
You can see that there is significant rock failure around the excavation. Observe however that the installation of the liners at the top of the tunnel have prevented extensive failure in this area, while at the bottom of the tunnel, where a liner has not been applied, the failure is severe. This is not too worrying since much of this failed rock will be excavated in the next stage.

Stage 3 shows further excavation and liner installation. Stage 4 completes the installation of the concrete liner. You will see that the addition of the concrete liner in stage 4 has virtually no effect on the stress or failure in the rock.

We can examine the role of the composite liner by plotting axial forces and bending moments.

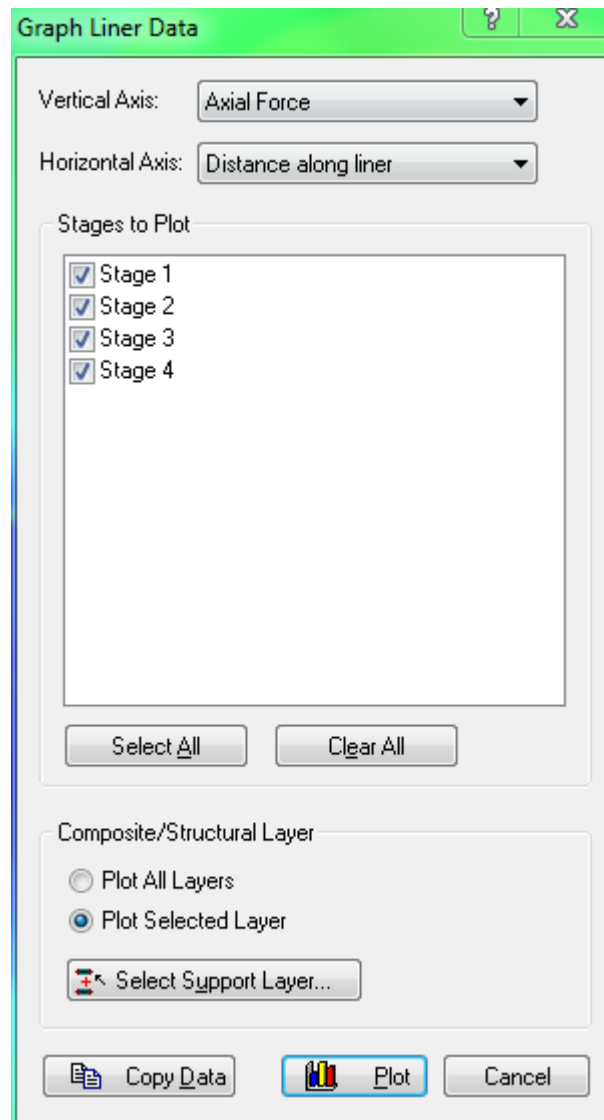


Click on the Graph Liner Data button. Click on the boundary of the tunnel and hit Enter. First we will look at the effect of the shotcrete only. So click on the **Select Support Layer** button. You will see under Liner that "Layer 1: Shotcrete" is selected.

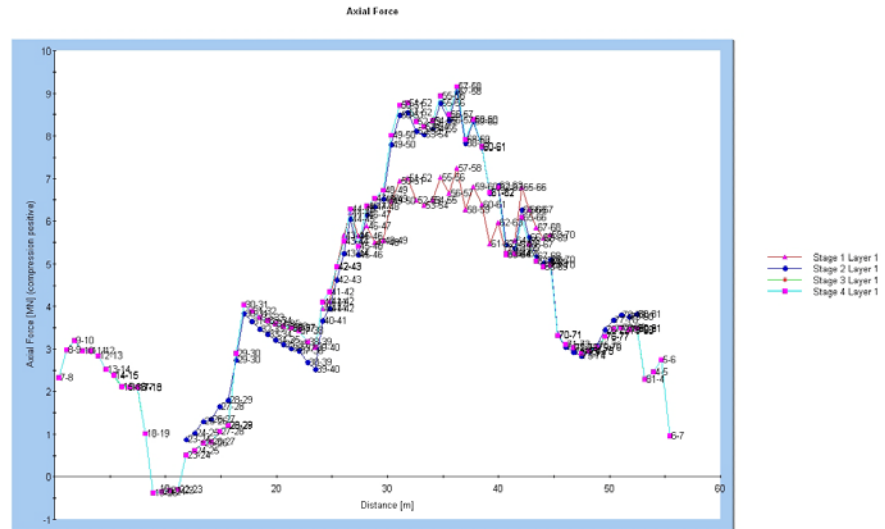


Since this is what we want, click OK to close the window.

Now select all stages in the Graph Liner Data dialog.



Click Create Plot. The graph should look like this.

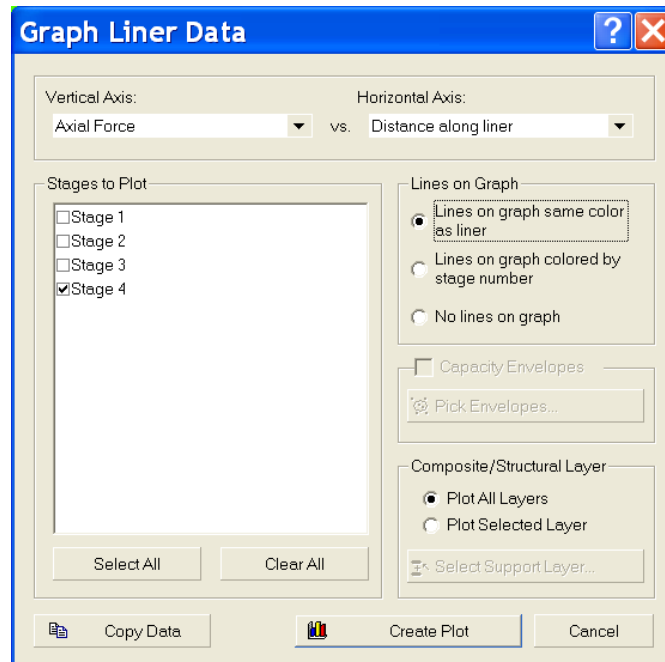


You will see that the axial force increases after Stage 1 at the top of the tunnel (between 30 m and 45 m). This reflects increased stress in this area as the tunnel excavation proceeds.

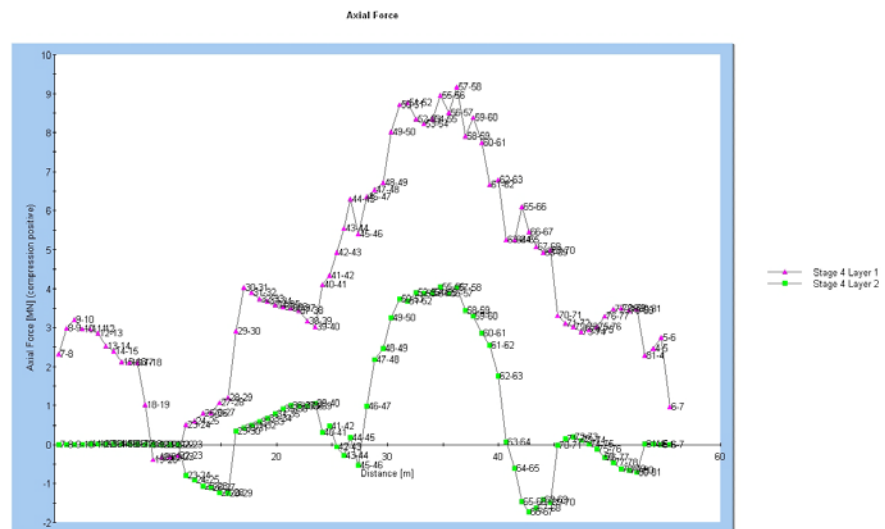
The numbers on the lines display the node numbers from the model so you can determine which points on the graph correspond to which sections of the tunnel boundary. Go back to the plot of the tunnel. You will see labels along the liner showing the node numbers.



We can also plot the effect of the two different liners on the same graph. Click on the Graph Liner Data button again and select the tunnel boundary. Hit Enter. Now under **Composite / Structural Layer** in the bottom right corner of the dialog, select **Plot All Layers**. Under **Lines on Graph** select **Lines on graph same colour as liner**. Turn on Stage 4 and turn off all of the other stages.



Click Create Plot. Your graph should look like this.

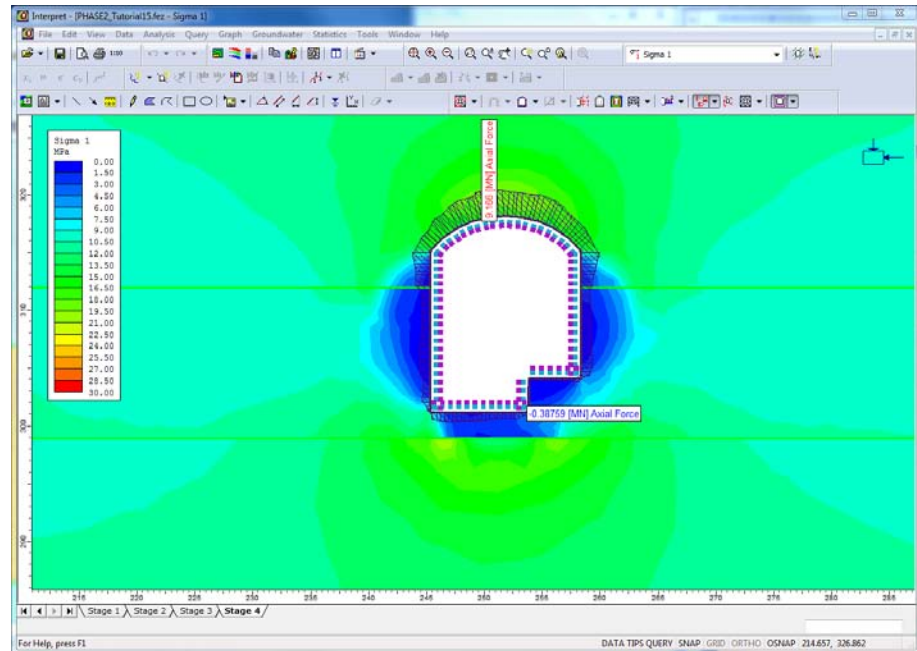


You can see that the shotcrete (Layer 1) is generally taking more load than the concrete. This is because the shotcrete was installed first. When the concrete is installed on top, it takes no load until further excavation occurs and stresses around the tunnel are redistributed. This is the reason that the concrete shows zero axial force along the bottom of the tunnel (the left side of the graph) – there is no further excavation after it is installed and therefore no load is shifted to the concrete support layer.

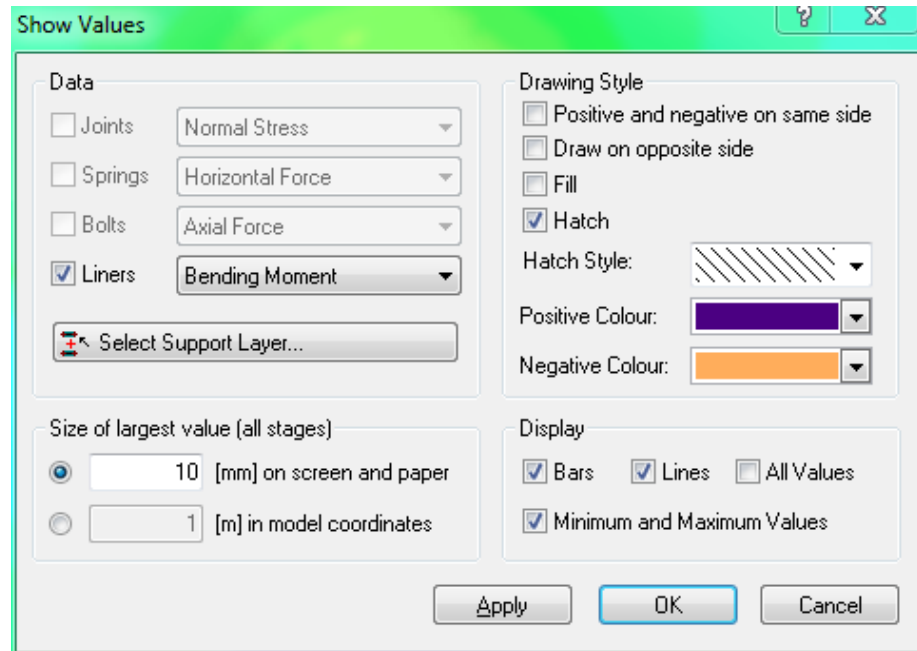
To make interpretation easier, you may want to graph the loads directly on the plot of the tunnel. Go back to the plot of the tunnel. Ensure you are looking at Stage 4. Turn off the Yielded Elements. Turn off the liner numbers by right clicking on the liner and clicking on Liner Numbers.



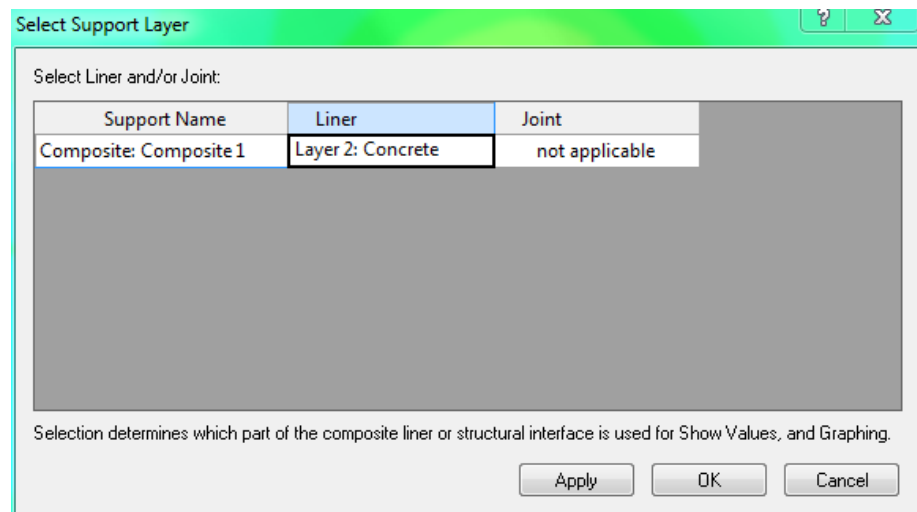
Click on the Show Values button. In the Show Values dialog, select the Liners checkbox, and make sure Axial Force is the data type. Select OK.



Minimum and maximum forces are shown by blue text and red text respectively. To change the options select Show Values again. Next to Liners, choose Bending Moment from the drop down menu as shown.



Now click on the Select Support Layer button. Under the Liner heading, click in the box and select “Layer 2: Concrete”.

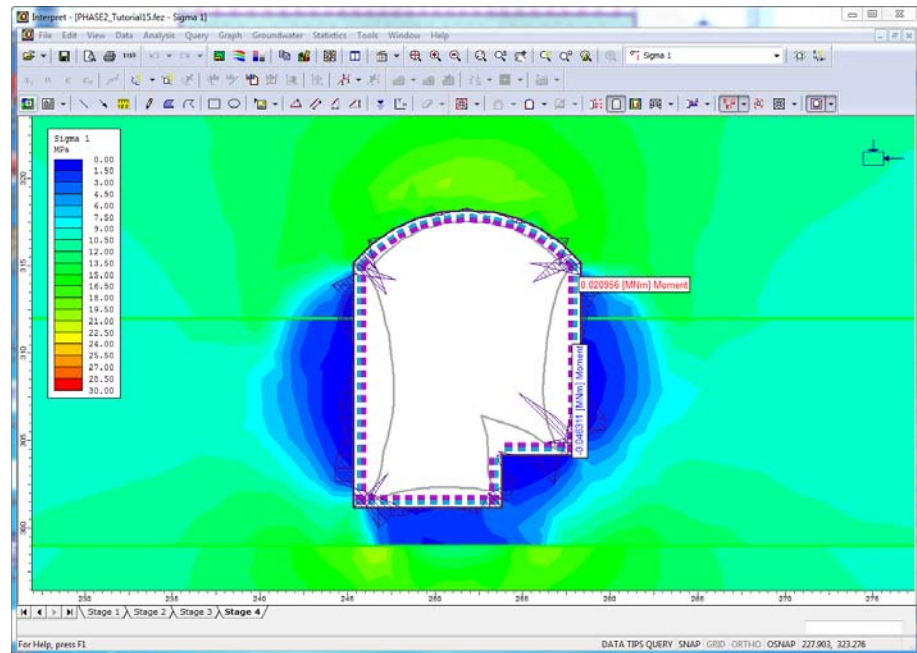


Click OK. Now click OK in the Show Values dialog.

TIP: you can also Show Values by right clicking on the liner and using the options in the Show Values sub-menu.



You should now see the bending moments in the concrete liner plotted around the tunnel. If you want, you can plot the exaggerated displacement of the tunnel by clicking on the Display Deformed Boundaries button. The plot should now look like this.



You can see the large bending moments at the top corners of the tunnel. You may want to turn off the text showing the minimum and maximum moments since they are obscuring some of the data. You can do this by going to the Show Values dialog and clearing the checkbox next to the Minimum and Maximum Values option.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.